

Grade 8
Unit 3
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

1. What is the value of 10^{-16} , written as a fraction?

$$\frac{1}{10,000,000,000,000,000}$$

2. Complete the statements.

The expression $\frac{10^{-17}}{10^{-5}}$ can be written can as 10^{-12} . Another way to write the expression $\frac{10^{-17}}{10^{-5}}$ using positive exponents only is $\frac{1}{10^{12}}$ or $\frac{10^5}{10^{17}}$.

3. Complete the table. Write each mathematical statement in an equivalent form by applying once or more laws of exponents. Then, write the equivalent decimal value.

Mathematical Statement	Value Written as a Power of 10	Decimal Equivalent
$(10^4) - 4 \times 10^{25}$	Not possible	-40,000,000,000,000,000,000,000
$(\frac{1}{10^{-5}})^3$	10^{15}	1,000,000,000,000,000
$\frac{10^{-14}}{10^{-9}} \cdot 10^6$	$10^{-5} \cdot 10^6 = 10$	10

- or $\frac{1}{1,000,000,000,000}$ as a power of 10.

10^{-12}

- [illegible]

10^{-27}

Using a calculator, find the value of each mathematical statement. Then, if possible, rewrite the value as a power of 10 or write 'not possible'.

Mathematical Statement	Value Found Using the Calculator	Value Written as a Power of 10
6. $(10^{-2})^5$	.0000000001	10^{-10}
7. $10^{-4} - 10^{-1}$	-0.0999	not possible
8. $\frac{1}{10^{-9}}$	1,000,000,000	10^9
9. $\frac{10^{-4}}{10^{-6}}$	100	10^2
10. $10^{-6} + 10^{-2}$	.010001	not possible